Deterministic Guardrails and Variance-Aware Evaluation for Multi-Agent Content Personalization

Anonymous Submission
Author and affiliation details are withheld in accordance with JCST's double-blind review process.

Abstract Large language models produce fluent marketing text, yet they routinely fabricate quantitative claims and drift outside the task they were asked to perform, and the field lacks settled practice for deciding whether an engineering fix has genuinely helped. This paper reports a system together with a measurement study. The system is a personalized content generator in which five narrow agents share a three-level user profile, and in which two interventions are deliberately deterministic rather than delegated to the model: a scope guard that redirects mismatched requests without any model call, and a grounding guard that flags claim-shaped figures absent from user-supplied data instead of rewriting them. Adding the guards raises the deterministic assertion suite from 102 to 105 passing checks with the model held fixed. Substituting a larger model and changing nothing else closes the three residual failures, establishing them as a capability limit rather than a prompt defect. The measurement study then asks how far such results can be trusted. Scoring identical outputs three times with a fixed automatic judge reveals a mean score spread of roughly half a point on a five-point scale, and re-running an unchanged configuration moves a per-agent mean by up to a third of a point. Against that noise floor, neither a seven-model comparison nor a paired test of per-agent model assignment shows a significant benefit, contradicting the intuition that specialized agents merit specialized models. Running the evaluation also exposed two defects in the evaluation apparatus itself.

Keywords content personalization, evaluation methodology, large language models, measurement reliability, multi-agent systems

1 Introduction

Large language models produce fluent prose across many domains, yet deploying them for personalized content generation exposes two problems that fluency does not solve. First, output must conform to a specific user's brand identity, tone, platform conventions and accumulated preferences, none of which the model knows. Second, and more consequentially for professional use, models fabricate specifics. Asked to write marketing copy, they volunteer percentages, currency amounts and multipliers that no one supplied, in prose confident enough that a reader has no signal to doubt it [1].

A natural architecture for the first problem is to decompose the task across narrow agents, each holding its own instructions and profile, coordinated by a router [2, 3]. A natural approach to the second is to instruct the model not to invent figures. We implemented both and found the second does not work: explicit prohibition in the system prompt did not prevent fabrication, a result consistent with broader findings that prompt-level constraints are unreliable safety mechanisms [4].

That observation motivated a design preference that runs through this work. Where a property can be checked deterministically, we check it in code rather than asking the model to respect it. Two mechanisms follow. A *scope guard* compares the evidence a request provides for the selected agent's task against the evidence for every sibling agent, and redirects the request when the mismatch is unambiguous, without a model call and therefore at no latency or token cost. A *grounding guard* extracts claim-shaped numeric patterns from generated copy and checks each against the set of figures the user actually provided, flagging those it cannot justify.

Having built interventions, we needed to know whether they helped. This is where the work turned from engineering to measurement. Deterministic assertions answer narrow questions well: did the agent hand off, did a budget allocation sum correctly, did the reply honour a hashtag limit. They say nothing about whether copy is any good. The standard remedy is an automatic judge [5, 6], and judges are known to carry biases: they favour their own generations [7], favour longer answers, and are sensitive to presentation order [8, 9].

What we found is that the practical obstacle is not bias but *variance*. Scoring identical outputs three times with a fixed judge at temperature zero produced a mean score spread of 0.51–0.54 on a five-point scale. Re-running an unchanged system configuration and re-judging moved one agent's mean by 0.37. Both figures are larger than every per-agent quality difference we were otherwise preparing to act on. Reported without that context, a difference of 0.3 reads as an improvement; measured against it, the difference is indistinguishable from noise.

This paper addresses four research questions.

- **RQ1.** Do deterministic scope and grounding guards measurably improve a personalized multi-agent content system, and which residual failures are attributable to model capability rather than system design?
- **RQ2.** How reliable is an automatic judge applied to this task, and what does its measured variance imply for the differences that may be claimed?
- **RQ3.** Among freely available models, how much does the choice of generator model affect output quality relative to that variance?
- **RQ4.** Does assigning a different model to each specialized agent improve quality over a single model serving all of them?

Our contributions are as follows.

1. A personalized multi-agent architecture in which scope enforcement and claim grounding are deterministic post-processing stages rather than prompt instructions, with measured effect (Subsection 5.1).
2. A reliability protocol for judge-based evaluation that reports score spread across repeated judgements alongside the scores, together with the measured noise floor for this task (Subsection 5.2).
3. A seven-model comparison and a paired test showing that per-agent model specialization yields no statistically significant benefit at this scale, with the negative result reported in full (Subsections 5.3 and 5.4).
4. Two defects in our own evaluation apparatus, surfaced by executing it: a rubric-induced collapse of a five-point scale to three

points, and a test case whose rubric anchor penalized correct behavior (Subsection 5.5). We report these because they proved more instructive than the system results they were meant to produce.

2 Related Work

2.1 Personalized Text Generation

Classical personalization relies on explicit user models or collaborative signals [10, 11]. For large language models, three families dominate. Fine-tuning adapts weights to a user or cohort, which is effective but costly and impractical per user. Retrieval-augmented generation [12] injects user-specific material at inference, and the LaMP benchmark [13] formalizes personalized generation as a retrieval-plus-generation task across several sub-problems. Prompt-level conditioning simply appends profile text, which is cheap and effective [14] but unstructured: profiles tend to accumulate as free text with no notion of which attributes apply to which task.

Our architecture differs in composing profiles from three nested scopes, global, domain and agent, so an attribute is declared once at the level where it is true and an agent receives exactly the union that applies to it. We are not aware of prior work that reports measured effects for scoped profile composition in a deployed multi-agent setting, though the mechanism itself is unsurprising.

2.2 Multi-Agent Systems Built on Language Models

Agent frameworks such as AutoGen [2], CrewAI [15] and LangGraph [3] coordinate specialized components built on the transformer architecture [16], drawing on reasoning-and-acting patterns [17], intermediate-step prompting [18], tool invocation [19] and self-critique [20]. These frameworks principally address *task decomposition*: splitting a complex objective across cooperating roles. Comparatively little attention is paid to what happens when a user addresses the wrong agent, which in a user-facing product is routine rather than exceptional. Our scope guard targets that case specifically, and we evaluate it as a first-class mechanism.

2.3 Hallucination and Guardrails

Hallucination in generated text is extensively surveyed [1]. Detection approaches include sampling-based self-consistency checks [21], atomic fact decomposition against references [22], and retrieval-grounded metrics [23]. Programmable guardrail systems such as NeMo Guardrails [4] place a controllable layer between user and model.

Our grounding guard is narrower than any of these, deliberately so. It does not assess truth; it assesses *provenance*. A percentage in generated marketing copy is flagged unless that figure appears in the user's request, attachment or profile. This is weaker than fact-checking and much cheaper, with no model call and no retrieval corpus, and it fits the deployment reality that only the user knows whether a claim about their product is true. Critically, the guard flags rather than rewrites: deleting a number mid-sentence yields text such as "reduce costs by up to", which is worse than the original.

2.4 Automatic Judges and Their Reliability

Using a strong model to score generated text is now standard [5, 6], validated against human preference at scale [24]. Known failure modes include self-preference, where an evaluator rates its own generations higher [7], position and verbosity effects [8], and broader cognitive biases documented systematically by Koo *et al.* [9]. Mitigations include swapping presentation order, ensembling multiple judges, and sampling multiple judgements in the spirit of self-consistency [25].

The gap we address is reporting practice rather than method. Judge scores are commonly published as point estimates, with bias acknowledged qualitatively and variance not quantified at all. We measure the variance for our task, find it large enough to subsume the differences we were about to claim, and argue that an explicit noise floor should accompany any judge-derived comparison. The observation that such evaluation pipelines are themselves under-tested is, to our knowledge, rarely reported with concrete instances; Subsection 5.5 supplies two.

3 System Design

3.1 Overview

The system serves five narrow marketing agents: a social-caption writer, a search-optimization agent, a campaign planner, and two agents whose deliverable is a rendered image, namely a social post image and an advertisement creative. Each is declared, not coded, as a specification carrying its instructions, profile field definitions, output kind and guard configuration; the runtime is shared. A broader earlier configuration of the same architecture spanned seven domains and 34 agents, which we use here only to argue that the design is not specific to five agents.

Requests traverse the eight-stage pipeline in Fig.1. Two stages are highlighted because they are the deterministic interventions this paper evaluates.

3.2 Three-Level Profile Composition

A user holds a *global* profile covering identity and communication style, a *domain* profile which for marketing carries brand name, industry, audience, tone, palette and prohibited terms, and an *agent* profile which for the caption writer carries platforms, target length and hashtag policy. At request time the three are composed into a context block, with the narrowest scope taking precedence. A brand tone is stated once and reaches all five agents; a hashtag policy reaches only the agent for which it is meaningful.

3.3 Deterministic Scope Guard

Let a be the selected agent and S the set of sibling agents. For a request r we compute a task-evidence score $\tau(x, r)$ for each agent x, accumulating matches of r against the task vocabulary declared by x. The request is redirected to $x^\star = \arg\max_{x \in S} \tau(x, r)$ when all of the following hold:

$$|r| \geq \lambda_{\min} \tag{1}$$

$$\tau(a, r) \leq \theta_{\text{own}} \tag{2}$$

$$\tau(x^\star, r) \geq \theta_{\text{other}} \tag{3}$$

$$\tau(x^\star, r) - \tau(a, r) \geq \delta \tag{4}$$

with $\lambda_{\min} = 15$ characters, $\theta_{\text{own}} = 1.5$, $\theta_{\text{other}} = 4.0$ and $\delta = 3.0$. The thresholds are deliberately conservative: a wrong redirect is more damaging to trust than a slightly off-topic answer, so the guard stays silent unless the mismatch is clear. Condition (1) suppresses the guard for greetings and acknowledgements, which carry no task evidence either way. Condition (2) requires that the selected agent be a genuinely poor fit rather than merely a worse one, and conditions (3) and (4) require that some sibling be both strongly indicated and clearly better. All four must hold, so a request that is ambiguous between two agents is answered by the one the user chose.

One detail proved essential. An early version scored agents using their full declared vocabulary, including their own *names*. A

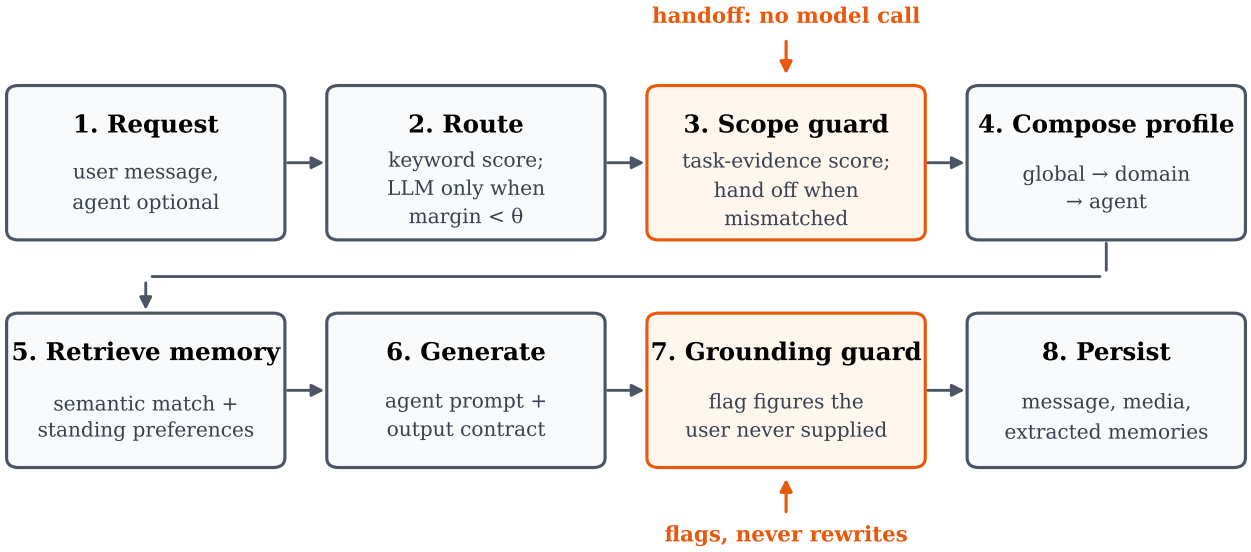


Fig. 1. The request pipeline. Stages 3 and 7 are deterministic: the scope guard redirects a mismatched request with no model call, and the grounding guard annotates unsupported figures without altering the generated text.

request such as "design an ad creative for the free trial" sent to the caption writer then scored highly for the caption writer, because the phrase contains words that appear in its own name, and the request was never redirected. Restricting τ to *task* evidence and excluding name matches fixed this. The router, which selects an agent when the user has not, continues to use the full score, since there a name match is a legitimate signal of intent.

3.4 Grounding Guard

Let P be a set of claim-shaped patterns: percentages, currency amounts, bare multipliers such as tenfold, and "up to" constructions, declared per agent. Let $N(r, \alpha, \pi)$ be the set of numeric tokens appearing in the request r, any attachment α, and the composed profile π. For generated text g, every match $m \in P$ whose numeric token $n(m)$ does not belong to $N(r, \alpha, \pi)$ is flagged. The reply is returned unchanged with an appended notice naming the unsupported figures.

Three refinements were required by observed false positives. *Cadence expressions* such as "three times per week" are schedules, not claims, and are excluded by a negative lookahead over temporal units. *Self-labelled proposals* such as a percentage annotated as a suggested figure are explicitly marked as proposals, which is behavior the campaign planner's instructions request, and flagging them penalized compliance. *Budget allocation* is legitimate for the planner: given a user-supplied total T, currency amounts no greater than T are permitted, enabled per agent so the exemption does not leak to agents that have no business allocating money.

The guard reads text only. When a user attaches an image, a figure the model read off that image is indistinguishable from one it invented, so the notice changes wording to say the check cannot see images rather than asserting fabrication.

4 Evaluation Methodology

4.1 Dataset

The dataset is 30 request cases held as declarative data, spanning twelve categories (Table 1). Categories were chosen to cover the failure modes we cared about rather than to sample naturally occurring traffic, a limitation we return to in Section 7. Five cases are marked as requiring a real model and are skipped under the deterministic composer, because that composer ignores agent instructions by construction and running them would say nothing about prompt quality.

Table 1. Composition of the Evaluation Dataset of 30 Request Cases

Category	n	Category	n
Ordinary	7	Grounding	4
Out of scope	6	Complex	3
Profile	2	Routing	2
Ambiguous	1	Attachment	1
Brand	1	Clarification	1
Memory	1	Platform	1

Note: Categories are disjoint. "Out of scope" and "Routing" cases produce a handoff rather than creative output and are therefore excluded from judge scoring.

4.2 Deterministic Assertions

Each case declares machine-checkable assertions: expected output kind, whether a scope handoff should occur and to which agent, rendered media dimensions, substrings that must or must not appear, hashtag and length limits, whether profile content reached the assembled prompt, and whether a stored memory was retrieved. A run yields 110 assertions over the 30 cases. Every run records the provider, model, temperature, token budget, embedding configuration and guard thresholds, and the comparison tool refuses to compare runs that differ in these respects.

4.3 Rubric and Judge

Qualitative dimensions are defined in a human-readable rubric of 32 dimensions across five groups, each with anchors at levels 1, 3 and 5. A case declares which dimensions apply to it; 22 of the 30 cases are scorable, the remaining eight being scope handoffs and routing decisions that produce no creative output.

The judge receives the brand context the run actually configured,

Table 2. Deterministic Suite Before and After the Guards, With Model and Settings Held Identical

	Baseline	With guards
Assertions passed	102/105	**105/108**
Errors	0	0
Degraded image specifications	0	0
Routing of "write ad copy"	Image agent	**Caption agent**
Supplied budget of 400	Not asserted	**Sums to 400**
Invented figures	Undetected	**Flagged**
Elapsed, 21 generating cases	1215 s	1322 s

Note: The guards add three assertions, which pass. Elapsed time rises by 9%, which is the cost of longer instructions.

the request, the output, and the anchors for that case's dimensions, and returns a score and a one-sentence justification per dimension at temperature zero. Two dimensions, legibility and brand fit, are properties of rendered pixels; since the judge receives only the image specification and alternative text, these are reported as unjudged rather than guessed. Layout legibility is instead checked deterministically across 27 size-and-payload combinations.

The rubric is parsed from the same prose document a human reader uses, so the machine-readable and human-readable versions cannot diverge. A consistency check asserts in both directions that every dimension a case declares exists in the rubric and every dimension defined is used by some case.

4.4 Reliability Protocol

The protocol below is what Subsection 5.2 quantifies, and what every comparison in Subsection 5.4 depends on.

The judge scores each case N times. We report the *median* as the score and the *spread*, that is the maximum minus the minimum, as the uncertainty. With $N = 1$ no spread is reported, since unknown is not the same as zero. Runs are comparable only when judge model, judge prompt version, rubric file and N all match, and the comparison tool enforces this. When the judge model equals the model that produced the output, the condition is flagged in the terminal, the stored results and the rendered report, because self-preference makes such a figure an upper bound rather than a measurement [7].

4.5 Reproducibility

All reported numbers derive from stored run artefacts rather than from transcription: every figure in this paper is generated by a script that reads those artefacts, so a figure cannot drift from the result it depicts. Hardware was a four-core mobile processor without a graphics accelerator. Local models ran under a self-hosted runtime; hosted models were reached through a vendor-neutral chat-completions interface. The source code, dataset and all run reports will be released publicly on acceptance; they are withheld during review to preserve anonymity.

5 Results

5.1 Effect of the Deterministic Guards

Table 2 reports the deterministic suite before and after the guards, holding the model and all settings fixed at a local three-billion-parameter model.

Three failures persisted, all on cases requiring the model to ask a clarifying question or to refrain from inventing figures. No prompt revision moved them. Substituting a larger hosted model, with every other setting unchanged, closed all three and reduced elapsed time from 1325 s to 88 s. We therefore attribute those residual failures

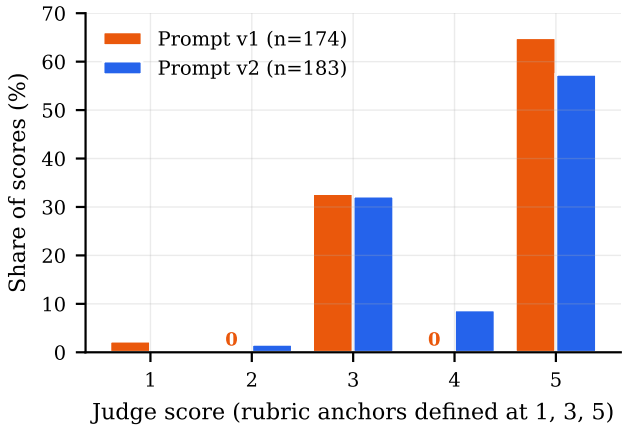


Fig. 2. Judge score distribution before and after the judge prompt named the intermediate scale levels. Under version 1 no score of 2 or 4 occurred in 174 judgements, because the rubric defines anchors only at 1, 3 and 5.

to model capability rather than system design, a distinction that deterministic assertions alone cannot draw, and that only a controlled model substitution could establish.

That substitution also exposed two false positives in our own grounding guard that the weaker model had never triggered, because it never produced the constructions involved: a posting cadence and a self-labelled proposal, both described in Subsection 3.4. Both were corrected, and the corrected guard is checked in both directions, so unlabelled claims are still caught.

5.2 Judge Reliability and a Rubric-Induced Scale Collapse

Judging the full suite with an independent model, not the generator, produced a mean of 4.23 out of 5 across 61 dimension scores. Taken alone this number is close to meaningless, which is the point of this subsection.

First, the scale. Across 174 individual judgements under our initial judge prompt, *every score was 1, 3 or 5*; levels 2 and 4 were never used (Fig.2). The cause was the rubric: its tables define anchors at 1, 3 and 5, and the judge treated those as the only admissible answers. A five-point instrument had silently become a three-point one. Naming the intermediate levels explicitly in the prompt restored them, with 10.4% of subsequent scores even-valued, and reduced the worst observed spread from 4 to 2.

Second, the variance. Scoring identical outputs three times at temperature zero gave 62%–69% exact agreement, a mean spread of 0.51–0.54 and a worst-case spread of 4 on a five-point scale. Re-running an unchanged configuration and re-judging moved one agent's mean by 0.37 with no change to the system at all. We adopt 0.5 as the noise floor for this task: differences below it are not results. An earlier screening pass used 0.33 and duly identified two per-agent winners that Subsection 5.4 shows to be spurious.

Scores are therefore comparable only within a fixed judge model, prompt version and repeat count, all of which we record with every stored result.

5.3 How Much Does the Generator Model Matter?

We evaluated every model available to us at no cost, seven of the twenty-two offered by the hosted provider, the remainder requiring paid credits, holding the dataset, thresholds and judge fixed and changing only the generator (Table 3, Fig.3).

Table 3. Seven-Model Comparison With Dataset, Thresholds and Judge Held Fixed

Model	Judge	Assertions	Time (s)	Errors
beam	4.16	108/110	142	0
gemma4:31b	4.15	109/110	**43**	0
gpt-oss:20b	3.89	105/110	113	0
nemotron-3-ultra	3.87	107/110	386	0
nemotron-3-super	3.80	107/110	731	0
gpt-oss:120b	3.52	103/110	83	0
nemotron-3-nano:30b	—	80/82	563	9

Note: gpt-oss:120b judged its own output, so its score is an upper bound. nemotron-3-nano:30b is excluded from the quality comparison: 9 of its 30 cases errored, so its mean describes an easier subset.

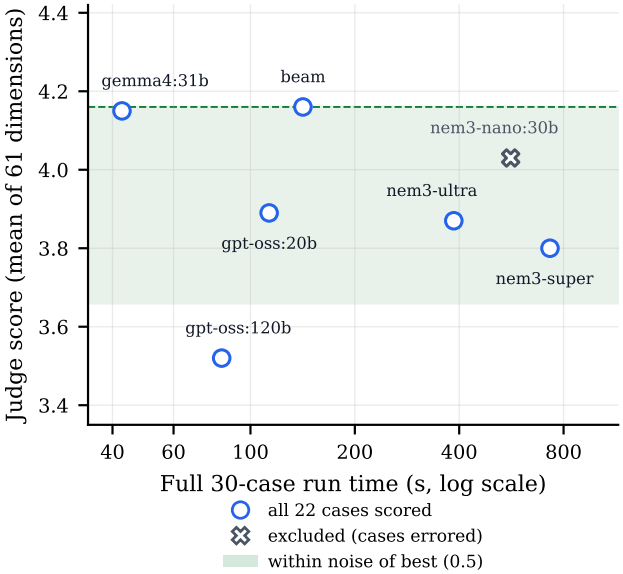


Fig. 3. Judge score against wall-clock run time for every freely available model. Five of six comparable models fall within the measured noise floor of the best, while run time varies by a factor of seventeen.

The leading two models differ by 0.01, far inside the noise floor, while differing by a factor of 3.3 in run time. Five of the six comparable models lie within 0.5 of the best. The practical conclusion is that for this task, selection among freely available models should be driven by latency and reliability, which are measured precisely, rather than by judge score, which is not.

Two incidental findings deserve note. The largest model tested scored *lowest* despite judging its own output, a condition that should have inflated it. And one model could not complete the suite at all, for a reason that turned out to be a defect in our own client (Subsection 5.6).

5.4 Per-Agent Model Specialization

A multi-agent architecture invites the hypothesis that each specialized agent should use the model best suited to it. The screening pass in Subsection 5.3 appeared to support this, identifying two agents whose scores improved by 0.45 and 0.33 under non-default models.

We tested the hypothesis directly. A mixed-fleet configuration pinned those two agents to their apparent best models; a control arm used one model throughout. Both arms were freshly generated and judged under the protocol of Subsection 4.4 with three repeats (Fig.4, Table 4).

Table 4. Paired Comparison of One Model Serving All Agents Against a Mixed Fleet

	One model	Mixed fleet
Overall judge mean	**4.28**	4.18
Assertions	109/110	109/110
Elapsed	**54 s**	201 s
Advertisement agent	**4.56**	4.33
Caption agent	3.80	**3.87**
Post-image agent	**4.25**	3.88

Note: The advertisement and caption agents are the two that were pinned to non-default models. The post-image agent used the same model in both arms, and its mean nonetheless moved by 0.37.

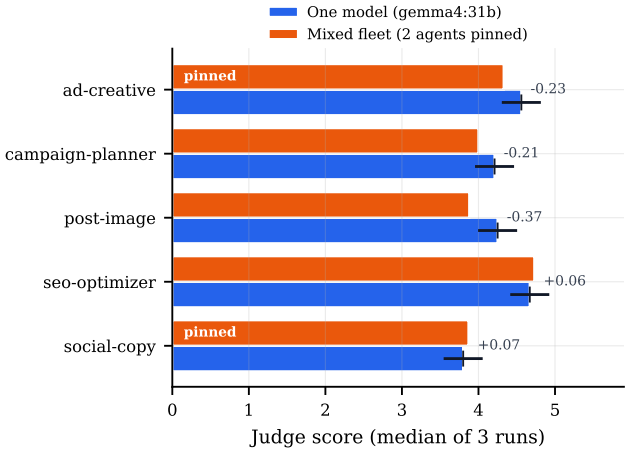


Fig. 4. Per-agent scores for both arms. Whiskers show the judge's own mean spread on the control arm. Every between-arm gap, including for the two pinned agents, lies inside that spread.

Of 61 paired dimension scores, 46 were unchanged. Among the 15 that moved, 7 favoured the mixed fleet and 8 the single model. A Wilcoxon signed-rank test [26] on the paired scores gives $W = 45.5$ with $n = 15$ and $p = 0.42$, exact and two-sided, with an effect size of $d = -0.12$: no significant difference and a negligible effect. We use a signed-rank test rather than a paired t-test because the scores are ordinal and discrete, and the exact distribution is tractable at this sample size. The mixed fleet was nonetheless 3.7 times slower.

The decisive observation is the last row of Table 4. The post-image agent used the *same* model in both arms and its mean still moved by 0.37. Generation at non-zero temperature plus judge disagreement produces per-agent movement of that magnitude with no configuration change whatsoever. The screening gaps of 0.33 and 0.45 were within that band, and did not survive confirmation; one of the two pinned agents became measurably worse.

We therefore reject the per-agent specialization hypothesis at this scale. We emphasise that this is a negative result about effect size relative to noise, not evidence that model choice never matters; a task with wider genuine quality separation could well reach the opposite conclusion.

5.5 Self-Audit: Defects in the Evaluation Apparatus

Two defects in the evaluation were found by running it, and both would have silently corrupted conclusions.

The scale collapse of Subsection 5.2 is the first. A rubric that reads correctly to a human restricted the judge to three of five levels across 174 judgements. Nothing in the pipeline would have surfaced this; it

Table 5. Latency and Cost on a Four-Core Mobile Processor Without a Graphics Accelerator

Operation	Local model	Hosted model
Text agent	28–53 s	1.3–3.2 s
Image agent, spec and render	28–64 s	1.4–2.4 s
Scope handoff	<1 ms	<1 ms
Full 30-case run	1325 s	49–88 s
Cost per run	—	0.0039 USD

Note: The scope handoff requires no model call, so a redirected request costs no tokens and returns effectively instantly regardless of backend.

was visible only on inspecting the score distribution, which we had no particular reason to plot.

The second was a self-contradictory test case. A case intended to test profile awareness configured the agent's content type as a landing page while the request asked for a blog post. The agent followed the explicit request and stated that it had done so, which is the correct precedence. The rubric anchor for that dimension rewarded adherence to the configured profile, and so scored correct behavior 1 out of 5, unanimously across three repeats. The defect lay in the rubric and the case, not the system. We corrected the anchor to state that an explicit request overrides a standing default, removed the contradiction from the case, and added a companion case that deliberately exercises the override.

Both incidents share a shape: the apparatus was internally consistent and externally wrong, and the error presented as a confident number.

5.6 A Client Defect Presenting as Provider Failure

During the model comparison, four of six models failed between 4 and 19 of 30 cases with what appeared to be provider outages. The cause was neither the provider nor the models. Models that spend completion tokens on hidden reasoning before answering can exhaust the token budget mid-thought, in which case the interface returns a success status with an empty message body and a finish reason of length. Our client treated an empty body as a provider failure. On one model, one request in six returned 700 completion tokens and zero characters of content.

Treating that specific signature, namely empty content with a length finish reason or a populated reasoning field with no answer, as a budget condition and retrying once with threefold headroom reduced five of six models to zero errors. The sixth remains unusable for this task: it produced 12 000 to 16 000 characters of reasoning and no answer even at a 4000-token ceiling, requiring 41 retries in a single run.

We report this because it affected comparability rather than merely robustness. Before the fix, each model errored on a different number of cases, so each model's score described a different subset of the dataset. Any ranking computed across those runs would have been an artefact of which cases happened to survive.

5.7 Latency and Cost

Table 5 contrasts a local three-billion-parameter model on commodity hardware with a hosted thirty-one-billion-parameter model. On a processor without a graphics accelerator, prompt evaluation dominates, so latency tracks prompt size as much as reply length.

6 Discussion

Prompt instructions are not a safety mechanism. Explicit prohibition did not stop figure fabrication; a deterministic post-processing check did. This agrees with the premise behind programmable guardrail systems [4] and suggests a simple design heuristic: if a property can be checked in code, check it in code, and reserve the model for what only a model can do.

Flagging beats rewriting for provenance. An automatic rewrite that removes an unsupported figure damages the sentence containing it, and the system cannot know whether the claim was true, since only the user can. Annotating preserves both the text and the user's authority over it.

Report the noise floor, not just the score. This is our main methodological claim. Our judge disagreed with itself by roughly half a point on a five-point scale, and total run-to-run movement reached 0.37 on an unchanged configuration. Differences of 0.3 are routinely presented as improvements in informal practice. Publishing a judge score without its variance invites exactly the error we made during screening and caught only by confirming. A noise floor is cheap to obtain, requiring only that the judge be run N times and the spread reported, and it changes which conclusions are available.

Architectural specialization does not imply model specialization. The appeal of giving each narrow agent its own model is strong and, at this scale, unsupported: no significant difference, a negligible effect size, 3.7 times the latency and more operational complexity. One model served five agents as well as five models did.

Evaluation code deserves the same scrutiny as system code. Both defects in Subsection 5.5 were in the measuring apparatus, and both produced plausible numbers. We found them only because we plotted a distribution and re-read a low score we did not believe.

6.1 Implications for Practitioners

Four findings translate directly into deployment practice, and we state them separately because each is cheap to adopt.

First, separate the two kinds of check. Deterministic assertions over machine-verifiable properties are fast, free and stable, and they should carry as much of the evaluation burden as possible; reserve judge scoring for properties that genuinely require reading. In our suite, 110 assertions cover routing, handoff, provenance, geometry, length and profile propagation, leaving 32 rubric dimensions for the judge.

Second, record the configuration with the result. Every stored run in this work carries its provider, model, temperature, token budget, embedding settings, guard thresholds and, after the defect in Subsection 5.6, its per-agent model assignments. The comparison tool refuses to compare runs that differ in any of these. This sounds bureaucratic until a mixed-model run is indistinguishable from a single-model run in the archive, which was briefly the case here.

Third, make the evaluator's uncertainty a first-class output. Reporting a median with its spread costs one extra field and N times the judge calls, in our case 213 s and no monetary cost, and it is the difference between a conclusion and a guess.

Fourth, prefer flagging to silent correction wherever the system cannot verify the property itself. The grounding guard annotates, the scope guard redirects and says why, and the empty-reply handler retries and then reports the specific cause. In each case the user retains the information needed to override the system, which matters more in content generation than a marginally cleaner output would.

7 Threats to Validity

Construct validity. Judge scores are a proxy for human judgement, not a measurement of quality. We did not conduct a human scoring pass, so we cannot report agreement between our judge and

human raters; this is the most important limitation of the work. Self-preference was controlled by using a judge different from the generator in all reported comparisons and by flagging the one case where it was not, but verbosity and leniency effects were not controlled: 36 of 61 median scores were 5s.

Internal validity. Agents generate at a temperature of 0.4, so arms differ in generated text as well as configuration; our paired comparison therefore measures the combined effect, and we report the control arm's own variance to bound it. The signed-rank test rests on 15 non-tied pairs, which is small, and a larger dataset could detect effects we cannot. Two of the scored dimensions were excluded as unjudgeable from text, so image agents are scored on fewer dimensions than text agents.

External validity. All results use a single synthetic brand in one industry, one language and one set of five agents. The 30 cases were authored to probe failure modes we anticipated, not sampled from production traffic, so category proportions do not reflect real usage. The free-model set is specific to one provider at one point in time, and model availability changes. The noise floor of 0.5 is a property of this task and judge, not a constant.

Conclusion validity. Our fourth research question reports a negative result; absence of a significant difference at this sample size is not evidence of absence of any effect. Our claim is bounded accordingly: per-agent specialization is unjustified *at this scale and for this task*, because any true effect is smaller than the variance we measured.

8 Conclusions

We presented a personalized multi-agent content generation system in which scope enforcement and claim grounding are deterministic rather than prompt-mediated, and a measurement study of what can honestly be concluded about such a system. The guards took the deterministic suite from 102 of 105 assertions to 110 of 110 once a capable model was used, and a controlled model substitution showed the residual failures to be a capability limit rather than a design defect.

The measurement results are the more transferable contribution. An automatic judge applied to this task disagreed with itself by a mean spread of 0.51–0.54 on a five-point scale, and an unchanged configuration moved a per-agent mean by 0.37. Against that floor, neither the seven-model comparison nor a paired comparison of per-agent model assignment showed a significant benefit, and the mixed fleet cost 3.7 times the latency. Running the evaluation also surfaced two defects in the evaluation itself, each of which had been producing confident and wrong numbers.

We conclude that judge-derived comparisons should be published with a measured noise floor, that evaluation apparatus warrants the same testing discipline as the system under test, and that architectural specialization should not be assumed to justify model specialization without a paired, repeated measurement.

Future work should establish agreement between automatic and human judgement for this task, extend the dataset beyond a single brand and language, and test whether the noise floor we measured is characteristic of marketing-copy evaluation or of our judge in particular.

References

[1] Ji Z, Lee N, Frieske R, Yu T, Su D, Xu Y, Ishii E, Bang Y, Madotto A, Fung P. Survey of hallucination in natural language generation. *ACM Computing Surveys*, 2023, 55(12): 1–38.

[2] Wu Q, Bansal G, Zhang J, Wu Y, Li B, Zhu E, Jiang L, Zhang X, Zhang S, Liu J, Awadallah A H, White R W, Burger D, Wang C. AutoGen: Enabling next-generation large language model applications via multi-agent conversation. arXiv:2308.08155, 2023.

[3] LangChain. LangGraph: Build resilient language agents as graphs. Technical documentation, 2024.

[4] Rebedea T, Dinu R, Sreedhar M, Parisien C, Cohen J. NeMo Guardrails: A toolkit for controllable and safe large language model applications with programmable rails. In *Proc. the 2023 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, Dec. 2023, pp.431–445.

[5] Zheng L, Chiang W L, Sheng Y, Zhuang S, Wu Z, Zhuang Y, Lin Z, Li Z, Li D, Xing E P, Zhang H, Gonzalez J E, Stoica I. Judging large language models as judges with MT-Bench and Chatbot Arena. In *Advances in Neural Information Processing Systems 36*, Dec. 2023.

[6] Liu Y, Iter D, Xu Y, Wang S, Xu R, Zhu C. G-Eval: Natural language generation evaluation using large language models with better human alignment. In *Proc. the 2023 Conference on Empirical Methods in Natural Language Processing*, Dec. 2023, pp.2511–2522.

[7] Panickssery A, Bowman S R, Feng S. Large language model evaluators recognize and favor their own generations. arXiv:2404.13076, 2024.

[8] Wang P, Li L, Chen L, Cai Z, Zhu D, Lin B, Cao Y, Liu Q, Liu T, Sui Z. Large language models are not fair evaluators. arXiv:2305.17926, 2023.

[9] Koo R, Lee M, Raheja V, Park J I, Kim Z M, Kang D. Benchmarking cognitive biases in large language models as evaluators. arXiv:2309.17012, 2023.

[10] Koren Y, Bell R, Volinsky C. Matrix factorization techniques for recommender systems. *Computer*, 2009, 42(8): 30–37.

[11] Zhang Y, Chen X. Explainable recommendation: A survey and new perspectives. *Foundations and Trends in Information Retrieval*, 2020, 14(1): 1–101.

[12] Lewis P, Perez E, Piktus A, Petroni F, Karpukhin V, Goyal N, Küttler H, Lewis M, Yih W T, Rocktäschel T, Riedel S, Kiela D. Retrieval-augmented generation for knowledge-intensive natural language processing tasks. In *Advances in Neural Information Processing Systems 33*, Dec. 2020, pp.9459–9474.

[13] Salemi A, Mysore S, Bendersky M, Zamani H. LaMP: When large language models meet personalization. arXiv:2304.11406, 2023.

[14] Brown T B, Mann B, Ryder N, Subbiah M, Kaplan J, Dhariwal P, Neelakantan A, Shyam P, Sastry G, Askell A *et al*. Language models are few-shot learners. In *Advances in Neural Information Processing Systems 33*, Dec. 2020, pp.1877–1901.

[15] Moura J. CrewAI: Framework for orchestrating role-playing autonomous agents. Software repository, 2024.

[16] Vaswani A, Shazeer N, Parmar N, Uszkoreit J, Jones L, Gomez A N, Kaiser L, Polosukhin I. Attention is all you need. In *Advances in Neural Information Processing Systems 30*, Dec. 2017, pp.5998–6008.

[17] Yao S, Zhao J, Yu D, Du N, Shafran I, Narasimhan K, Cao Y. ReAct: Synergizing reasoning and acting in language models. In *Proc. the 11th International Conference on Learning Representations*, May 2023.

[18] Wei J, Wang X, Schuurmans D, Bosma M, Ichter B, Xia F, Chi E, Le Q, Zhou D. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems 35*, Dec. 2022, pp.24824–24837.

[19] Schick T, Dwivedi-Yu J, Dessì R, Raileanu R, Lomeli M, Hambro E, Zettlemoyer L, Cancedda N, Scialom T. Toolformer: Language models can teach themselves to use tools. In *Advances in Neural Information Processing Systems 36*, Dec. 2023.

[20] Shinn N, Cassano F, Gopinath A, Narasimhan K, Yao S. Reflexion: Language agents with verbal reinforcement learning. In *Advances in Neural Information Processing Systems 36*, Dec. 2023.

[21] Manakul P, Liusie A, Gales M J F. SelfCheckGPT: Zero-resource black-box hallucination detection for generative large language models. In *Proc. the 2023 Conference on Empirical Methods in Natural Language Processing*, Dec. 2023, pp.9004–9017.

[22] Min S, Krishna K, Lyu X, Lewis M, Yih W T, Koh P W, Iyyer M, Zettlemoyer L, Hajishirzi H. FActScore: Fine-grained atomic evaluation of factual precision in long form text generation. In *Proc. the 2023 Conference on Empirical Methods in Natural Language Processing*, Dec. 2023, pp.12076–12100.

[23] Es S, James J, Espinosa-Anke L, Schockaert S. RAGAS: Automated evaluation of retrieval augmented generation. In *Proc. the 18th Conference of the European Chapter of the Association for Computational Linguistics: System Demonstrations*, Mar. 2024, pp.150–158.

[24] Chiang W L, Zheng L, Sheng Y, Angelopoulos A N, Li T, Li D, Zhang H, Zhu B, Jordan M, Gonzalez J E, Stoica I. Chatbot Arena: An open platform for evaluating large language models by human preference. In *Proc. the 41st International Conference on Machine Learning*, Jul. 2024.

[25] Wang X, Wei J, Schuurmans D, Le Q, Chi E, Narang S, Chowdhery A, Zhou D. Self-consistency improves chain of thought reasoning in language models. In *Proc. the 11th International Conference on Learning Representations*, May 2023.

[26] Wilcoxon F. Individual comparisons by ranking methods. *Biometrics Bulletin*, 1945, 1(6): 80–83.